

EXPERIMENT 27:

Demonstrate Infrastructure as a Service (IaaS) by establishing the remote connection, launch the created VM image and run in your desktop..

Aim

To demonstrate **Infrastructure as a Service (IaaS)** by establishing a remote connection to the previously created **Azure Virtual Machine**, launching the VM, and accessing the cloud-based operating system from the desktop.

Procedure

1. Open the **Microsoft Azure Portal** and sign in using the Azure account.
2. Navigate to **Virtual Machines** and select the previously created virtual machine.
3. Open the **Overview** page and verify that the VM status is **Running**. If it is stopped, click **Start** and wait until the VM starts successfully.
4. Note the **Public IP address** of the virtual machine from the Overview page.
5. For a Linux VM, open **PowerShell/Command Prompt** on the desktop and establish an SSH connection using the VM's username and public IP address.
6. Authenticate using the SSH private key configured during VM creation. After successful authentication, the remote Ubuntu terminal will be displayed on the desktop.
7. Execute basic Linux commands such as `hostname`, `ls`, and `uname -a` to verify that the commands are running on the Azure VM.
8. Take screenshots of the Azure VM Overview and the successful remote terminal connection as evidence of the implementation.

IMPLEMENTATION:

```
swap:                                0B                0B                0B
azureuser@IaaS-Test-VM:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G   1.6G   27G   6% /
tmpfs            422M    0   422M   0% /dev/shm
tmpfs            169M  956K   168M   1% /run
tmpfs            5.0M    0    5.0M   0% /run/lock
efivarfs         128K   38K   86K   31% /sys/firmware/efi/efivars
/dev/sda16       881M   64M   756M   8% /boot
/dev/sda15       105M   6.2M   99M   6% /boot/efi
tmpfs            85M   12K   85M   1% /run/user/1000
azureuser@IaaS-Test-VM:~$ cd f1
-bash: cd: f1: No such file or directory
azureuser@IaaS-Test-VM:~$ cat f1
cat: f1: No such file or directory
azureuser@IaaS-Test-VM:~$ mkdir cloud
azureuser@IaaS-Test-VM:~$ cd cloud
azureuser@IaaS-Test-VM:~/cloud$ cat>f1
currently running vm^Z
[1]+  Stopped                  cat > f1
azureuser@IaaS-Test-VM:~/cloud$ cat f1
azureuser@IaaS-Test-VM:~/cloud$
azureuser@IaaS-Test-VM:~/cloud$ cat>f2
writing file2 content
^Z
[2]+  Stopped                  cat > f2
azureuser@IaaS-Test-VM:~/cloud$ ls
f1 f2
azureuser@IaaS-Test-VM:~/cloud$ cd f1
-bash: cd: f1: Not a directory
azureuser@IaaS-Test-VM:~/cloud$ cat f1
azureuser@IaaS-Test-VM:~/cloud$ cat f2
writing file2 content
azureuser@IaaS-Test-VM:~/cloud$
```

Result:

The Azure Virtual Machine was successfully launched and accessed remotely from the desktop using an SSH connection. The remote Linux environment was displayed and basic commands were executed successfully, demonstrating the use of Infrastructure as a Service (IaaS) through remote access to cloud-based computing resources.